phone directory that contains tables of telephone numbers and corresponding physical persons' names. For each separate physical network, separate ARP table or binding table is constructed. IP address entries in the table bear the same network ID for all nodes. As separate ARP table is required for separate physical networks, network ID (prefix of IP address)—being the same for all nodes—is omitted from ARP table to save memory space, like the telephone directory of Kolkata may not require code 033 in the table of directory.

Main advantage of table look-up is simplicity in operation. Once an IP address is known, a search of the table will resolve the physical hardware address of the corresponding node, host or computer. When a network has less than a dozen hosts or nodes, a sequential search will suffice. For networks having a higher number of hosts, nodes or computers, hashing or direct index search would be a better approach. Table look-up technique is basically used in WAN.

Closed-form computation technique of address resolution is another technique used in ARP. It is meant for networks that use the technologies of configurable addressing. In configurable addressing, physical addresses are not static. In closed-form computation, a computational, mathematical or relational function exists between the IP address and the corresponding hardware address. For example, a class B network with address 156.57.<u>00001100.00000001</u> (host ID is underlined for illustration) is a configurable network. As and when computers are added to this network, physical addresses to the computers may be assigned such that 1s complement of the host's ID in IP address becomes the corresponding physical address. Thus, the physical address, in the stated example, will be 00000000.00000001 (the 1s complement of host ID: 1111110011.1111110). Thus, a mathematical rule is for resolving address.

The message passing technique of address resolution is mostly used for address resolution in LAN. Nodes in LAN may work with look-up tables, too. In that case, look-up tables are known as ARP cache. To resolve any address, nodes first search their respective ARP cache. When ARP cache fails to supply the necessary resolution, message passing is evoked. The node then sends a broadcasted ARP request to all other stations of its network to find the link address of the target or destination IP address. Any station that recognises the target IP address sends a reply to the inquiring node. The reply contains the physical address of the target IP address. This technique is illustrated in Fig. 3. Source node S sends ARP request and node D sends ARP reply.

Reverse address resolution protocol (RARP). Given a logical IP, the protocol used to derive its physical address is what is called ARP. Whereas, given a physical address, the protocol used to derive its logical address is RARP. RARP is the reverse of ARP. RARP is not so commonly used as ARP because most stations or computers know their IP address. All of these have a hard disk to store the same.

RARP is required for stations having no hard disk-like terminals or disk-less workstations. To run RARP, a RARP server is required in the network. A RARP server maintains a mapping table like that of a look-up table used in ARP. In ARP, all nodes are at par. But in RARP, the service is client-server oriented.

The inquiring node sends a broadcast RARP request. RARP server recognises the request and searches its database to find the protocol address for the given physical address of the inquiring node. The server then sends a RARP reply to the inquiring node. When a node sends RARP request, the destination hardware address will be the hardware address of the RARP server. The destination protocol address will be a broadcast address, which is net ID plus all 1s in the field of host ID. The message format will be that of the servicing node, both for ARP and RARP. If the service network is Ethernet, the message format will be of Ethernet frame. EFY

The article is dedicated to Robert E Kahan and Vint Cerf, who developed protocol suite TCP/IP in 1978